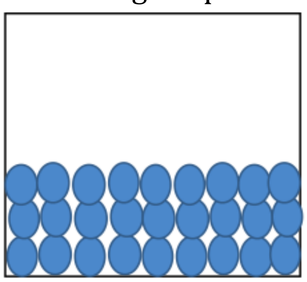
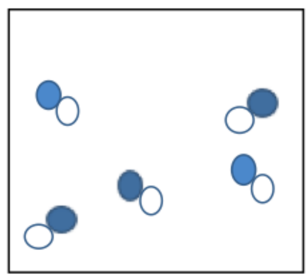
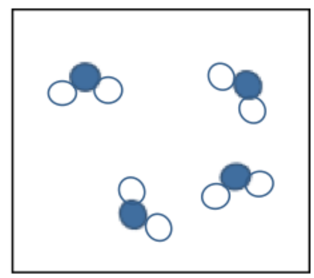
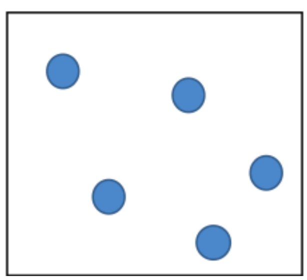
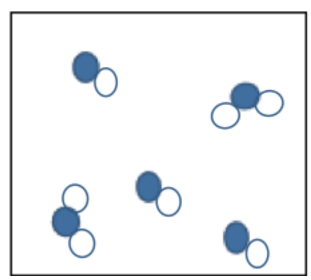
1. The statements below pertain to the development of the theory of combustion by the French chemist Lavoisier in the 18th century. Which statement is matched correctly with the appropriate step of the scientific method?
   A) Observation: Combustion of a metal in a closed container ceases after a length of time.
   B) Hypothesis: Oxygen is the gas that combines with a substance during its combustion.
   C) Experiment designed to test hypothesis: A metal is burned in a closed container and the change in mass of the solid and volume of the gas is measured.
   D) None of the statements are correctly matched to steps of the scientific method.
   E) All of the statements are correctly matched to the steps of the scientific method.

2. Which image depicts a mixture?

A)

B)

C)

D)

E)


3. A sample of element X contains 100 atoms with a mass of 12.00 amu and 10 atoms with a mass of 14.00 amu. Calculate the average atomic mass of element X.

   A) 12.18 amu
   B) 13.00 amu
   C) 13.40 amu
   D) 13.82 amu
   E) 15.20 amu

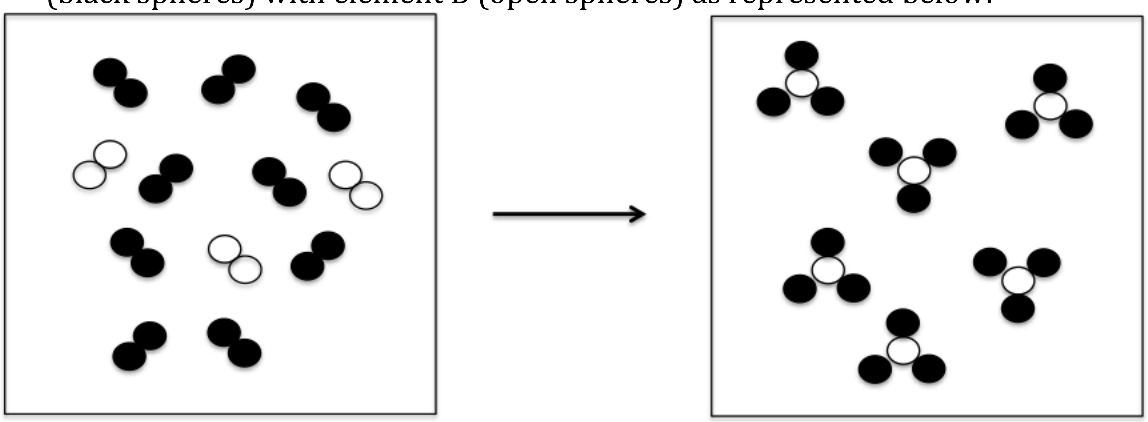
4. What mass of octane, C_8H_{18}, is required to provide 1.50×10^{24} carbon atoms?

   A) 0.00351 g
   B) 0.0281 g
   C) 35.6 g
   D) 285 g
   E) 2.14×10^{24} g

5. What is the symbol for an ion that has 92 protons, 146 neutrons, and 86 electrons?

   A) ${}^{146}_{86}\mathrm{Rn}$
   B) ${}^{146}\mathrm{Rn}^{6-}$
   C) ${}^{238}\mathrm{U}$
   D) ${}^{238}_{92}\mathrm{U}^{6+}$
   E) ${}^{146}\mathrm{U}^{6-}$

6. Which answer *best* represents the balanced equation for the reaction of element A (black spheres) with element B (open spheres) as represented below:

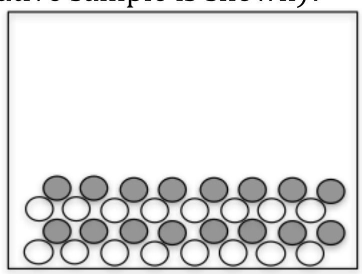


   A) $9\ A_2 + 3\ B_2 \rightarrow 6\ A_3B$
   B) $18\ A + 6\ B \rightarrow 6\ A_3B$
   C) $3\ A_2 + B_2 \rightarrow 2\ A_3B$
   D) $3A_2 + 3B_2 \rightarrow 3A_3B$
   E) $9\ B_2 + 3\ A_2 \rightarrow 6\ AB_3$

7. Which compound incorporates *only* covalent bonding?

A) $CaBr_2$
B) NH_4Cl
C) $Fe(NO_3)_2$
D) C_3H_8
E) NaI

8. Which *correct* chemical name could be applied to the ionic solid shown below (assume that a representative sample is shown)?



A) Nitrogen monoxide
B) Copper(I) chloride
C) Magnesium bromide
D) Calcium phosphate
E) Sodium(I) oxide

9. Combustion analysis of a given hydrocarbon that is used in the production of mothballs produced 17.60 g of CO_2 and 2.88 g of H_2O. Determine the empirical formula for the hydrocarbon.

A) CH
B) C_2H_5
C) C_5H_4
D) C_5H_8
E) C_6H

10. How many grams of N_2 are required to produce 15.5 g of NH_3 by the following chemical reaction if the percentage yield of the reaction is 85.0% and H_2 is present in excess? Molar masses (all in g/mol): H_2 = 2.02, N_2 = 28.02, and NH_3 = 17.04

$$3\,H_2(g) + N_2(g) \rightarrow 2\,NH_3(g)$$

A) 10.8 g
B) 12.7 g
C) 15.0 g
D) 21.7 g
E) 25.5. g

11. Cisplatin, [$Pt(NH_3)_2Cl_2$], a compound used in cancer treatment, is prepared according to the chemical reaction:

$$K_2PtCl_4 + 2\,NH_3 \rightarrow 2\,KCl + Pt(NH_3)_2Cl_2$$

What mass of which reagent remains when 55.8 g K_2PtCl_4 and 35.6 g NH_3 react? You may assume 100% yield.

Molar masses (in g/mol): K_2PtCl_4 = 415.08, NH_3 = 17.04, KCl = 74.55, $Pt(NH_3)_2Cl_2$ = 300.06

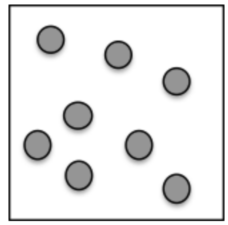
A) 15.4 g NH_3
B) 17.8 g NH_3
C) 20.2 g K_2PtCl_4
D) 31.0 g NH_3
E) 40.2 g K_2PtCl_4

12. Which statement is true regarding dissociation and dissolution?

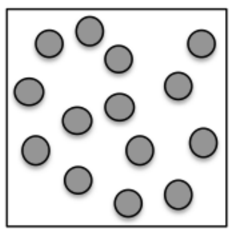
A) Nonelectrolytes do not dissociate or dissolve.
B) Weak electrolytes dissociate but do not dissolve.
C) Strong electrolytes dissolve and dissociate.
D) Weak acids dissolve and dissociate completely.
E) Dissolution and dissociation are two words that mean the same thing.

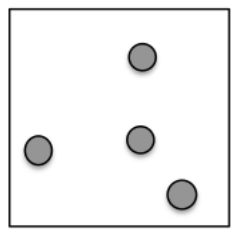
13. A solution is made by dissolving $AgNO_3$, $Ca(NO_3)_2$, and $Zn(NO_3)_2$. What substances could you add sequentially (in the order listed) to leave only Ca^{2+} ions in solution?
A) NaBr, NaCl
B) Na_3PO_4, NaCl,
C) NaI, Na_2S
D) Na_2CO_3, NaOH
E) Na_2CO_3, NaI

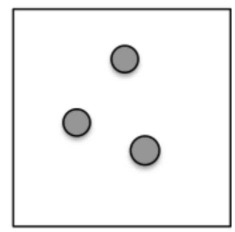
14. The image below represents a small volume within 25 mL of an aqueous solution of a covalent compound. Water molecules have been omitted for clarity.

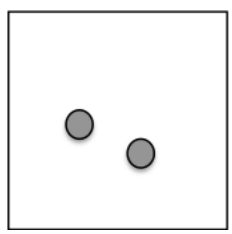


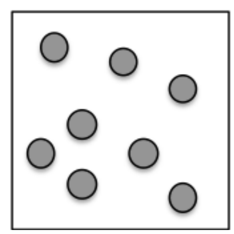
Which image below best represents the same volume of solution after an additional 25 mL of water are added?

A) 

B) 

C) 

D) 

E) 

15. The pressure exerted on a sample of a fixed amount of gas is doubled at constant temperature, and then the temperature of the gas in Kelvin is doubled at constant pressure. What is the final volume of the gas?

A) The final volume of the gas is twice the initial volume.
B) The final volume of the gas is four times the initial volume.
C) The final volume of the gas is one-half the initial volume.
D) The final volume of the gas is one-fourth the initial volume.
E) The final volume of the gas is the same as the initial volume.

16. Which of the following occupies 44.8 L at STP?

A) 32.00 g O_2
B) 56.04 g N_2
C) 2.02 g He
D) 35.45 g Cl_2
E) 39.95 g Ar

17. A mixture of He, Ne, and Ar gases has a total pressure of 662 mmHg under a specific set of conditions. If the partial pressure of He is 341 mmHg and the partial pressure of Ne is 112 mmHg, then what is the mole fraction of Ar, X_{Ar}, in the mixture?

A) $X_{Ar} = 0.0113$
B) $X_{Ar} = 0.316$
C) $X_{Ar} = 0.449$
D) $X_{Ar} = 0.684$
E) $X_{Ar} = 0.989$

18. An unknown gas effuses at a rate that is 0.4937 times that of oxygen gas (at the same temperature). The unknown gas is likely:
A) NH_3
B) Ar
C) H_2
D) Kr
E) Xe

19. Which statement is true regarding the following samples of gas?

I. O_2 at 298 K
II. O_2 at 344 K
III. He at 344 K

A) Samples I and II have the same kinetic energy but different most probable speeds.
B) Sample III has the greatest kinetic energy.
C) Samples I and II have the same kinetic energy.
D) Sample III has the greatest most probable speed.
E) Samples II has the lowest most probably speed.

20. In a given system, expanding gases do 451 J of work while 325 J of heat are lost to the surroundings. What is the change in internal energy for the system?

    A) +126 J
    B) -126 J
    C) +776 J
    D) -776 J
    E) -451 J

21. Which form of the exam do you have?

    A) B)